Robot Project Rules and Deadlines

1 Introduction

The objective of the "Cube Craze" robot project is to design and build a robot that moves cubes around an arena in a fun competition where you face off against your classmates. The goal of each match is to gather more cubes than your opponent's robot by the end of a one minute period (this objective has changed slightly from last year, where the goal was to bring as many cubes as possible to your side of the board). You can see photos and a highlight video from the 2019 competition hereLinks to an external site.. Students will compete in teams of 3, so we will have approximately 62 teams. You will be provided with a kit of stock parts, but you will also have access to fabrication facilities and be able to order parts from additional vendors (with a limited budget). This document outlines the rules of gameplay, team creation, robot constraints, guidelines for components and manufacturing, competition logistics, deliverables, and grading.

2 Timeline

Item/Deliverable	Due date/time	Points	Notes
Team signup	Friday, March 27th, 11:59pm	n/a	Team signup spreadsheet available on Canvas
Team contract	Tuesday, April 7th, 11:59pm	5	Team contract template available on Canvas
Parts order #1 (McMaster, Digikey, Pololu, Adafruit)	Monday, April 13th, 11:59pm	n/a	Order form will be available on Canvas
Milestone 1 (design pitch)	Friday, April 17th, end of open lab hours	10	See Milestones section
Milestone 2 (mobility)	Friday, April 17th, end of open lab hours	10	See Milestones section
Parts order #2 (McMaster, Pololu, Adafruit)	Monday, April 20th, 11:59pm	n/a	Order form will be available on Canvas
Milestone 3 (color sensing and border detection)	Friday, April 24th, end of open lab hours	10	See Milestones section
Parts order #3 (McMaster only)	Monday, April 27th, 11:59pm	n/a	Order form will be available on Canvas
Milestone 4 (cube clearing)	Friday, May 1st, end of open lab hours	10	See Milestones section
Final competition!	Tuesday, May 5th, 9am-2pm (approximate)	10	See Competition section
Final report	Wednesday, May 13th, 12:00pm (noon)	40	Rubric will be available on Canvas (group assignment, one report per group)

Item/Deliverable	Due date/time	Points	Notes
Individual portfolio	Wednesday, May 13th, 12:00pm (noon)	0	Important: this assignment is mandatory and must be completed in order to re
Peer evaluation	Wednesday, May 13th, 12:00pm (noon)	5	Peer evaluation form will be available on Canvas

3 Competition Rules and Overview

On Tuesday, May 5th, all teams will face off in a competition against the other groups. We will start out with a round-robin style tournament, where each team will compete against each other team in smaller group (not the entire class). Winners of each round robin group will advance to an elimination-style bracket tournament. The winning team from this bracket will face off against a robot built by ASML and the second-place team will go up against a robot built by the TAs. This section outlines the individual match rules, the playing field, and the scoring system.

3.1 Gameplay

Each match of the competition will adhere to the following rules:

Each match will be exactly one minute long.

A team member will start out by holding their robot in place on the board. A TA will count down "3, 2, 1, GO!" at which time the team member will either press the reset button on the Arduino or plug in the 9V battery on the Arduino.

The TA will have a 60-second countdown timer. At the end of the match, the TA will count down "5, 4, 3, 2, 1, STOP!" at which point team members will immediately immobilize their robots by grabbing them, holding them in place, and disconnecting/turning off power (either the 4xAA battery pack powering the motors or the 9V battery powering the Arduino).

The TAs act as referees/judges regarding false starts/stops, somebody moving/affecting the cubes when grabbing the robot, etc. A match may be restarted at the sole discretion of the TA, but all decisions are final. This is not the NFL, we do not have instant replay.

The team with the highest score (as outlined in Section 3.3) at the end of one minute wins the match.

Robots may implement any strategy the team sees fit as long as it does not violate any of the rules in any section of this document.

Besides starting and stopping the robot, the robot must be fully autonomous for the duration of the match. There will be no human intervention (no touching the robot to fix something or get it unstuck, no remote control, etc.) at any point during the match, except for TA intervention to ensure safety of spectators or the playing field.

Each side will start with 10 cubes (20 total), placed randomly in the middle third of the field by the TAs (see Figure 1).

Cubes and robots may leave and re-enter the field. There is no "out of bounds."

Each robot must start in the back of their zone, touching the black border, and centered horizontally on the field (you cannot start in a corner). The starting position will always be the same, however the team member can choose the robot's staring orientation (i.e. you can choose to face toward the middle of the field, toward a corner, backward, or at some other angle).

The TA will decide which side of the field (blue or yellow) each robot starts on.

Only one team member from each team may enter the roped-off playing area. This team member is responsible for starting and stopping the robot at signaled times.

Modifications to the robot during the course of competition - including but not limited to uploading new code, attaching a different sensor, toggling a switch once you know what color you start on, attaching new parts to the robot, etc. - are not allowed. Mechanical repairs needed to restore the robot to an operating state (re-connecting loose wires, taping on a part that fell off, etc.) are allowed if they can be done quickly between matches. We can't pause the entire competition for 30 minutes so you can go fix your robot.

3.2 Playing Field

The Cube Craze playing field is made from a 4'x4' sheet of 3/4" plywood covered with glossy self-adhesive vinyl. It is split into two 3.5'x1.75' colored zones (blue and yellow) with a 3" black border around the entire field. The playing cubes are 1" wooden blocks of varying color. Practice fields and cubes will be available during open lab hours. You can only use the practice equipment if a TA is present. Do not step on or otherwise damage the practice fields. Damaging the fields is disrespectful to the lab staff who put a lot of effort into making them. Students who ignore warnings about caring for the boards will have points deducted from their project grade. Repeat offenders will be disqualified from the competition.

competition board with dimensions

3.3 Scoring

The objective is for your robot to gather as many cubes as possible within one minute. The robot that has gathered the most cubes wins the match. So, what does it mean to "gather" a cube?

When viewed from directly above, cubes must be completely inside the bounding perimeter of your robot. This perimeter can be an irregular shape (it does not need to be a perfect circle or square etc.).

This means you cannot, for example, just cover the edges of your robot in loops of duct tape and drive around so the cubes stick to the tape. The result would be that the cubes are outside your robot's perimeter.

The "perimeter" must make a useful contribution toward gathering the cubes. You cannot, for example, just put an umbrella on top of your robot and use that to say that cubes below the umbrella are "inside the perimeter." The umbrella is not a mechanical part that actually helps gather the cubes. Robot design should follow the spirit of the competition (the point is to gather the cubes) - any attempts at finding further loopholes or circumventing this rule may be banned from the competition at the discretion of the course staff. Check with the TAs or Dr. Finio if you are not sure about your design.

You do not have to "pick up" the cubes. In other words, it is OK if the cubes are within the perimeter of your robot, but still resting on the competition board. It is also OK to pick up the cubes.

If a cube somehow ends the match inside the perimeter of both robots (e.g. the robots are interlocked/overlapping somehow), that cube does not count for either team.

Matches that result in a tie during the round robin portion of the competition will be recorded as a tie. Ties in the single-elimination bracket will result in a rematch until one robot wins.

robot-perimeter-cubes.png

4 Teams

You are allowed to choose your own group of three students. Note that depending on the exact number of students enrolled in the class, we may have one or two groups of two students. Teams of four are not allowed. A spreadsheet for team signup will be posted on Canvas (see deadlines at the top of this page). Each team will also be required to generate a team contract (template will be provided on Canvas). This contract is intended to help you develop a mutually agreed-upon set of rules for working on a team and help you avoid conflicts.

We realize that many students in the class know each other and want to work with their friends. We also have a lot of students from other departments or who don't know anyone, and may feel uncomfortable finding a team. In that case, there are two options:

You are welcome to create your own public post on Ed Discussion if you are looking to join a group, or if you already have a group of two and are looking for one more person.

If you would like to be placed on a team, please make a private Ed Discussion post. The course staff will monitor these and will form teams as needed.

You should review the team contract document and think things over before forming your team. For example, it's probably not a good idea to make a team of people with totally different schedules because it will be hard to find a time to meet. The contract worksheet lists other things you should consider when forming a team.

5 Robot Constraints

Each team must build one robot from the materials provided in your "lab bin" (this is different from the "individual kit" you have), with a budget of $40 for additional materials (not including shipping - orders will be placed in bulk for the class). The following section outlines the constraints and considerations you must factor into your design.

5.1 Dimensions

At the beginning of each match the robot must fit inside an 8"x8" square with no height limit. However, after the match starts, the robot may extend beyond its starting configuration but at any time must fit inside an imaginary 12" diameter cylinder (no height limit). Items dropped (i.e. purposefully detached from the robot) do not need to fit inside the 12" diameter cylinder.

5.2 Code

All code must run real-time on the Arduino UNO provided in your kit. No other microcontrollers, computers, wireless connections to smartphones, etc. are allowed. This also means you cannot pre-train a machine learning model on another computer - all computation must be done by the Arduino during the match. All code must be written in C using registers, not Arduino commands. Arduino commands are not allowed (pinMode(), digitalRead(), digitalWrite(), analogWrite(), micros(), map(), pulseIn(), third-party libraries for sensors, etc. - this is not an exhaustive list), with the limited exception of: the serial library for debugging purposes, analogRead() since this was not taught with registers, and the delay() command (which is similar to _delay_ms() but will accept a variable as an argument instead of a constant). If you aren't sure whether something you are doing is C or an Arduino command, ask a TA or on Ed Discussion before the competition. TAs will check your code for Arduino commands when you check in on the morning of the competition. Robots that use Arduino commands will not be allowed to compete.

5.3 Strategy

Each robot should employ a well thought out strategy, which includes algorithm and mechanical design. Note that since there are more than 40 teams and the competition has been done before, we understand that many teams may have similar designs or strategies - this is OK. However, you should be able to answer the question "Why did you go with your approach?"

5.4 Motors

Each robot must use the L9110H H-bridges provided in the kit to drive the motors. The wheel-driving motors must be servos provided in the kit (Parallax Continuous Rotation Servo #900-00008). Note that these servos have been modified to 2-wire operation and had the internal H-bridge removed, so they behave like a regular DC motor that is controlled with an external H-bridge. Additional motors can be purchased for other purposes (e.g. controlling arms/appendages) but cannot be used to drive the wheels.

5.5 Power

Electrical power is limited to four AA batteries and one 9V battery. No additional batteries or other sources of electrical power (e.g. a capacitor that was pre-charged from another power source) are allowed.

5.6 Reusability

At the end of the year the robot projects will be disassembled so parts can be re-used next year. Any modifications that cause permanent damage or make a part unusable in the future are not allowed. Some examples of prohibited modifications are listed below. If you are not sure about something, ask a TA or on Ed Discussion.

No physical modifications or soldering to major electronic components (the Arduino, servos, sonar/color/QTI sensors etc.). Soldering/trimming wires of minor components (resistors, capacitors, etc.) is OK.

No excessive use of glue or tape that makes it difficult to disassemble the robot or permanently gunks up any reusable parts.

No modifications to the chassis such as drilling additional holes.

5.7 Safety

This is not Battlebots. Your goal is to collect the most cubes, not to destroy the other robot. There is no protective cage or shield between spectators and the competition area. Therefore, safety of team members, course staff, and spectators is of utmost importance. If a team's robot is deemed to be unsafe it will be immediately removed from the competition and will forfeit remaining matches.

The following items are prohibited, but this is not an exhaustive list (someone always manages to think of a creative exception). If you have an idea that you're not sure about, please ask. If the answer to "Could your robot permanently damage another robot, the competition board, or injure a nearby human?" is "yes," then the answer to "Can I do ______?" is "no."

No chemical, biological, or nuclear weapons.

No blades (knives, saws, etc.).

No fire.

No dangerous potential energy storage (very stiff compressed springs, gas canisters, etc).

No projectiles that could damage another robot or injure a person. Items launched/dropped from the robot at low velocities are OK.

Nothing that could cause electrical damage to the other robot even if it does not cause physical damage (water, static electricity, etc.).

Nothing else that could injure a spectator.

Nothing else that could damage another robot such that it would be unable to continue in the competition. Temporarily stopping/disabling another robot (pushing it off the arena, tricking its sensors etc.) is OK.

Nothing that could damage or affect the surface finish of the competition field, including scratches, scuffs, paint/markers, residue from tape, etc.

6 Equipment, Purchasing, and Fabrication

Your team will be provided with a stock kit of parts. You will also have a $40 budget (not including shipping) available for fabrication and purchasing additional materials.

6.1 Provided Components

Each group will be provided with the following "free" components in your kit (they do not count toward your budget). These components should already be in your lab bin. If you are missing any components or any of them are not working, please let a TA know and we will get you a replacement (remember to go through proper debugging steps before you assume a part is not working).

Color sensor (1)

QTI sensor (1) - one additional sensor available for free (ask TAs)

Modified servo motor for driving wheels (2)

Positional micro servo (1)

Continuous micro servo (1)

Breadboard (1)

Ultrasonic sensor (1)

3-wire extension (4)

Red LED (2)

Green LED (2)

9V battery cap (1)

Robot chassis (1)

Robot wheels (2)

M-M 3-pin headers (5)

Rubber band robot tires (2)

L9110H H-bridge (2)

220 ohm resistor (2)

Nylon ball & cotter pin (1)

6.2 Free Components

The following items are free (they do not count toward your budget) while supplies last:

Wires

Standoffs and screws

Resistors, capacitors, and LEDs

Logic gates and transistors

Ask a TA to get more of these items. 3D printed color sensor mounts are also available to all teams (one per team), ask a TA to acquire one.

6.3 Batteries

Each team will be provided with one initial set of batteries for robot development (four AA's and one 9V). One additional set of batteries can be acquired for free during the testing period leading up to the competition (ask a TA in lab). Each team will receive a new set of batteries on the morning of the competition. Teams advancing to the finals (final two in the single-elimination bracket) will receive another new set of batteries.

6.4 Part Orders

The teaching staff will place part orders from vendors as listed in the table at the top of this page. Student orders are due at the dates/times listed in the table and will not be placed if they are not submitted on time. Note: due to the longer lead times from certain vendors, some of them are only available on the first order date.

Teams must fill out the order form available on Canvas to order parts. Make sure to look at the availability of the item you are purchasing - ensure there is no lead time / back orders!

Purchased parts may not exceed $40 total. This does not include shipping, but does include any items that are purchased but not used on the final robot.

All purchasing must be done through official vendors. Students are not allowed to spend their own money on the project.

Any scavenged/dumpster-dive/"laying around" parts must be assessed at their equivalent new purchase price and included in the budget.

Additional lab supplies may be purchased at reduced cost: QTI sensors ($2), IR sensors ($2), sonar sensors ($5), modified servos ($3), color sensors ($5), micro servos ($3), micro breadboards ($0.50), half breadboards ($1), continuous rotation servo with h-bridge intact ($3), set of 2 larger wheels ($4 total, $2 per wheel)

Recycled stock used in the machine shop will also count towards the $40 budget. See document on Canvas for pricing information.

Purely decorative items that do not affect the functionality of the robot in any way do not count toward your budget. You must supply these items yourself (you cannot order decorative items through the lab). Items that affect the functionality of the robot in any manner - e.g. are load-bearing, add a non-negligible amount of weight, or are intended to affect the other robot's sensors by reflecting/absorbing light or sound in a specific manner, etc. - must count towards your budget and may be purchased through the lab.

Examples: a small decoration/piece of paper or googly eyes do not count towards the budget (left). A large/heavy object counts towards the budget, even if it is primarily decorative (right).

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6.5 Working in the Mechatronics Lab

Starting the Monday after spring break, we will switch to an "open lab" format. However, this does not mean you can come whenever you want - we can only fit 12 groups in the lab at a time. Teams who wish to work in the lab must sign up for specific time slots. A schedule and signup spreadsheet will be posted on Canvas.

Teams may sign up for 3 slots per week.

In order to reserve a slot, enter your group number and the netID of one person from your group who will serve as a contact in the appropriate cell of the Google sheet.

Starting 30 minutes before the beginning of the first slot for the day, any slots that are still open will be filled on a first-come/first-served basis. These extra slots do not count towards a team's 3 slots per week limit. A team that wishes to use these extra slots must sign up for the available slot - do not just show up without having entered your group's info in the spreadsheet.

If you sign up and then decide to change or no longer need the slot, remember to delete your group from that time in the spreadsheet. It is not fair to other groups if you sign up for a slot and then don't show up.

If no one from your group has shown up 15 minutes into the scheduled slot, your group forfeits the slot. The TAs will delete your group's info so another group can fill the slot.

If your team tries to sign up for more than 3 slots per week (excluding same day slots per the third bullet above), your extra slots will be deleted.

Make sure to not mistakenly edit another group's entries in the spreadsheet. Course staff can see the spreadsheet edit history. If you intentionally try to steal another group's slot you will lose the slot, and at the instructor's discretion, possibly face additional penalties such as a grade reduction for the final project.

6.6 Working Out of the Lab

Teams may take their robot and lab bin outside of lab to work at home.

Due to the noise, mess, and disruption to another students, you can not work on this project in classrooms/conference rooms other than the Mechatronics lab and Upson 256.

To ensure fairness and equal access to resources for all teams, working in any access-restricted build space, including the ELL, private research labs, clubs/makerspaces, labs for other classes, or project team spaces, is strictly prohibited. At the discretion of the instructor, teams caught working in prohibited spaces may be banned from the competition and/or receive a penalty on their final project grade.

Tools you already have from your individual kits (multimeter, oscilloscope, screwdriver, pliers, wire strippers) can be used to work on the project outside of lab.

Tools from the lab (everything in the black toolboxes, the big multimeters, soldering irons, etc.) must remain in the lab. These tools must be put back on the benches/in the black toolboxes so other students can access them. Do not put them in your blue lab bins.

6.7 Fabrication

6.7.1 Machine Shop

Students will have access to the Emerson Machine Shop for working with sheet metal ($5 per 12x12 sheet, different thicknesses and metals available - ask in the shop). No machining will be allowed (mills, lathes, CNC, etc.). Time slots for machine shop availability will be TBD (likely M-F, 1:00-5:00 pm).

Signups for Emerson slots will be through a Google form on Canvas.

All teams using the machine shop must have their work approved by a TA.

All groups must check in and have their work approved by TBD (MAE Teaching Support) before starting work.

There will be no MAE 3780 work in the machine shop outside of open shop hours (no work during project team-specific hours).

6.7.2 RPL

Teams may also use the RPL for laser cutting and (3D printing TBD). A document with order information will be on Canvas asap.

6.8 Penalties

Lab items (big multimeters, hand tools from toolboxes) that are found in team bins will result in a 5% penalty on the final project grade. Double penalty for repeat offenses, quadruple penalty for taking items out of the lab.

Teams may only be in the lab if a TA is present. Teams that are in the lab without a TA will automatically forfeit the competition.

Teams that violate the rules about using restricted-access spaces to work on the project will automatically forfeit the competition.

7 Competition Day

On May 5th, all teams will compete against one another in a round-robin competition followed by a single-elimination tournament. Everyone will get the opportunity to play multiple matches against different opponents. The winners of the round-robin groups will move on to the single-elimination bracket. The following section includes an overview of the competition and competition-day logistics.

7.1 Overview

The competition will consist of a round robin competition (approximately 8 groups of 5-6 robots each) followed by a single elimination bracket. The top two teams in each round robin group will advance to the single elimination bracket.

Initial seeding in the tournament will be based on how quickly groups finish and submit the project milestones.

Matches will be decided in one round, except for the finals, which will be played between the final two competitors and will be best 2 out of 3.

The top two robot from the single elimination bracket will face off against boss robots built by ASML or the TAs.

In between rounds, each team should be ready to compete again in two minutes. We will have a projector showing which teams are currently competing and "on deck" for each of the six competition boards. A team that is a no-show for their match automatically forfeits that match.

7.2 Logistics

The competition will take place in Duffield Atrium on May 5th from 10:00am-1:00pm (end time approximate depending on how long the tournament takes). At least one member from your team must be present at all times in order to operate the robot. To ensure that this is possible, we are requiring that at least one member of the team is available at all times between 10am-1pm on May 5th. All team members are encouraged to attend the entire competition if possible, but you are not expected or encouraged to skip other classes for it.

7.2.1 Check-In

All teams are required to go through check-in the morning of the competition. Teams will sign up for check-in time slots using a spreadsheet available on Canvas. This will include a full inspection of the robot, including a size and code check and safety inspection. Teams will be required to demonstrate that their robot fits inside the 8"x8" starting square and never exceeds the 12" diameter cylinder.

Teams must upload their final code to the robot in lab while a TA watches and checks the code to make sure it does not use any Arduino commands. Robots that use Arduino commands will not be allowed to compete. Once the final code is uploaded, no further changes to the code are allowed and the team must turn in their USB cable. Upon successful completion of check-in, each team will be given a new set of batteries.

After checking in, teams should make sure they are in the Duffield Atrium by 9:30am with their robot. We will have a microphone and speakers set up in the room - await further instruction from the course staff.

7.2.2 Returning Equipment

After the competition, each team should return their robot and lab bin to Upson 264. If you haven't already, remember to take some pictures of your robot before you return it, as you will need them for the final report. Students that fail to return their kit (robot and everything provided by the lab) at the end of the final project will be charged a $500 replacement fee.

8 Milestones

Each team is required to complete several deliverables, or "milestones," to ensure progress toward a competition-ready robot. Teams may take as many attempts as needed to complete each milestone (except for the design pitch, which only happens once). You will not be penalized for failed attempts as long as you complete the milestone prior to its deadline. Late completion of a milestone will result in a 25% per day penalty for that milestone (homework slip days can NOT be used for project milestones). In order to complete a milestone, your team must demonstrate the functionality to a TA during lab (you can't do it at home and take a video of it).

The amount of effort required to complete each milestone is not uniform - they build on each other. While the milestones are designed to aid you in preparing a competition-ready robot, be sure to develop the other aspects of your robot in parallel with the milestones. Blindly following the milestone due dates will not be enough to get you through the competition.

8.1 Seeding

Teams will be ranked going into competition day based on when they completed the milestones (the sooner the milestones are completed, the higher the team will be ranked). Teams will be awarded 1 point for finishing the milestone first, 2 points for finishing second, and so on. Teams' total points will be the sum over all four milestones. Teams with the lowest total number of points will be seeded higher. This ranking will affect which round robin group you are placed in. The seeding will also determine the match pairings in the bracket if there is a tie. Note: the "points" here only affect seeding, they are not related to your grade.

In the single elimination bracket, seeding will be done by looking at the record during the round robin stage. If two teams have the same record coming out of round robin, then the team with the larger accumulated win margin will be seeded higher. If the teams are still tied, the higher initial seed will be seeded higher in the single elimination bracket.

8.2 Milestone 1: Design Pitch

Teams will give a 5 minute presentation to a TA with an overview of the mechanical design, program flowchart, and strategy for their robot. The presentation may take any form (slides on a computer, printed slides, poster, etc.) but must have some sort of visual aid. The rubric for this milestone will be available on Canvas.

8.3 Milestone 2: Mobility

Teams must demonstrate to a TA that their robot can move forward and backward specified distances, and make both left and right turns. To accomplish this, the robot must, in this order (see Figure 3):

Drive forward 1 foot

Turn right 90 degrees

Drive forward 1 foot

Turn left 90 degrees

Drive forward 6 inches

Drive backward 1.5 feet

Turn left 90 degrees

Drive forward 1 foot, returning to the starting position

Stop

While the robot does not need to move precisely the distances specified nor return to the exact position it started, it should perform all the actions listed above (all distances/angles are approximate, we aren't going to get out a protractor to check if your robot actually turned 89 degrees). If you are waiting on ordered parts to build a customized chassis or wheels, you must complete this milestone with the given chassis and wheels prior to the deadline. Completion of this task prior to the deadline will result in full credit for this milestone.

robot path for milestone 1

8.4 Milestone 3: Color Sensing and Border Detection

Teams must demonstrate to a TA that their robot can detect the current color of the robot’s location and the edges of the board. To accomplish this, the team’s robot must drive on a narrow a blue and yellow track, immediately stop when it hits the other color, turn 180 degrees, drive to the back of the starting color, and stop again. The robot must remain on the board at all times. The TA will randomly choose which color the robot starts on, the orientation of the robot and how far away the robot is from the other color. The robot must be able to independently determine which color it is on (the team will not be allowed to change code, flip a switch, etc. once they know which color the robot starts on).

The TAs are aware that a robot's sensors may not be located at the "front" of the robot and will judge the robot's reaction accordingly. Completion of the task prior to the milestone deadline will result in full credit for this milestone.

milestone 3

8.5 Milestone 4: Cube Clearing

Teams must demonstrate to a TA that their robot can gather cubes. At the beginning of the milestone, there will be 10 cubes on each side of the board, randomly scattered within the middle third of the field. To complete the milestone, the robot must gather at least 5 cubes within one minute. The TA will randomly choose the color the robot starts on. The robot will start centered at the back of the board (touching the black line), but the team may choose their desired starting orientation, much like the actual competition.

If the team’s robot successfully clears the cubes as specified above prior to this milestone’s deadline, the team will receive full credit for this milestone.

milestone 4